

Optimism and Pessimism Detection in Tweets using Label Engineering with Traditional and Deep Learning Models

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Abstract

Optimism and pessimism detection in social media can support well-being applications and help monitor collective attitudes. This project studies tweet-level optimism versus pessimism identification using SemEval-2018 Task 1 E-c data and a second Twitter dataset with sentiment and emotion scores. Since the SemEval data is multi-label and many tweets have neither optimism nor pessimism annotated, we design three label strategies that combine explicit labels with an aggregated emotion score derived from nine basic emotions. We compare two preprocessing regimes and five model families: Logistic Regression, linear SVM, TextCNN, DistilBERT and BERTweet. On the SemEval development set, BERTweet reaches a macro-F1 of 0.9005 under the percentile-based label strategy. We also evaluate cross-dataset generalization on the Dell tweets dataset with engineered labels and report strong transfer for transformer models.

1. Introduction

Optimism and pessimism represent a broader psychological stance that extends beyond traditional sentiment analysis. While sentiment classifiers capture polarity (positive/negative), they often fail to distinguish the nuanced attitudes underlying optimistic and pessimistic viewpoints [Caragea et al., 2018]. This distinction is particularly valuable for mental health applications and collective mood monitoring, where understanding whether users project hope or concern about the future provides deeper psychological insights than sentiment alone.

In this work, we address tweet-level optimism versus pessimism detection as a binary classification task. Our approach tackles the practical challenge of building reliable labels from multi-label emotion datasets – a common obstacle in sentiment and emotion analysis. We pursue four complementary objectives: (1) designing label strategies that combine explicit annotations with emotion scores to handle multi-label data; (2) comparing traditional models, neural networks and transformers to identify the most effective architectures; (3) evaluating gener-

alization to a second dataset; (4) providing explainability and error analysis to understand the models’ behavior and limitations.

By bridging label engineering with diverse model families, we aim to demonstrate how careful data preprocessing and label design can enhance classification performance.

2. Related Work

Caragea et al. [2018] explored optimism and pessimism detection on Twitter using deep learning, demonstrating that neural models outperform traditional classifiers. They built their dataset using a dual-threshold approach on multi-annotator scores ranging from -3 to $+3$, employing both strict settings (scores above or below 0) and selective settings (scores ≤ -1 or ≥ 1) to ensure clearer class separation. Importantly, they showed that optimism detection differs from sentiment analysis, as polarity alone cannot fully capture future-oriented psychological stances.

Cobeli et al. [2022] more recently studied optimism detection through a multi-task knowledge

distillation framework. By incorporating linguistic analysis and auxiliary signals from sentiment and hate speech detection, they suggested that targeted modeling decisions and external knowledge transfer can substantially boost performance on this task.

For our implementation, we use standard text classification approaches as baselines: linear models with TF-IDF, TextCNN [Kim, 2014], and pretrained transformers, including BERT [Devlin et al., 2019], DistilBERT [Sanh et al., 2019], and BERTweet [Nguyen et al., 2020]. This diverse set allows us to assess how model complexity and label engineering strategies affect performance on optimism and pessimism detection.

3. Methodology

3.1. Datasets and Exploratory Data Analysis

We use the **SemEval-2018 Task 1 E-c** dataset [Mohammad et al., 2018] as our primary source. This dataset is multi-label by design, which complicates binary optimism-pessimism classification. As shown in Figure 1, optimism and pessimism are underrepresented compared to core emotions like joy and anger. The joint label distribution (Figure 2) reveals that over 60% of tweets lack explicit optimism or pessimism labels, motivating our label engineering approach.

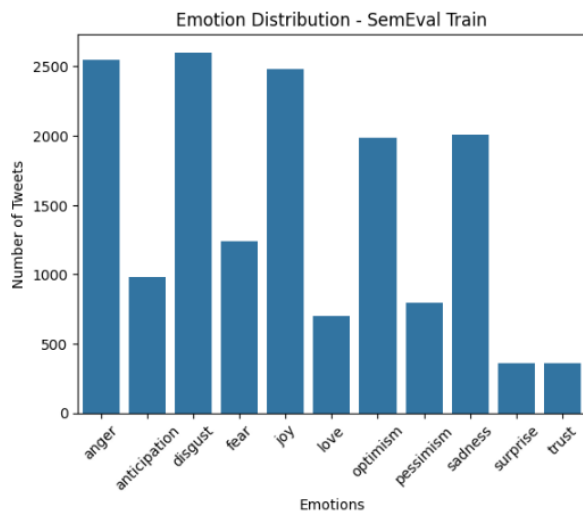


Figure 1: Emotion distribution in the SemEval-2018 training set.

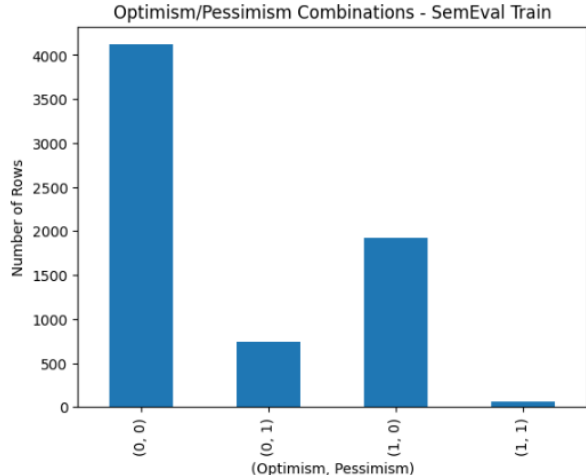


Figure 2: Joint distribution of (optimism, pessimism) labels in SemEval-2018 training set. Counts: (0,0) = 4118, (1,0) = 1925, (0,1) = 736 and (1,1) = 59.

For cross-dataset evaluation, we employ the **Dell Sentiment & Emotions Labelled Tweets** dataset [Ankitkumar2635, 2022], comprising approximately 25,000 brand mentions collected during the first three quarters of 2022. This corpus provides a realistic test of generalization, as customer interactions differ substantially from the general-domain SemEval data.

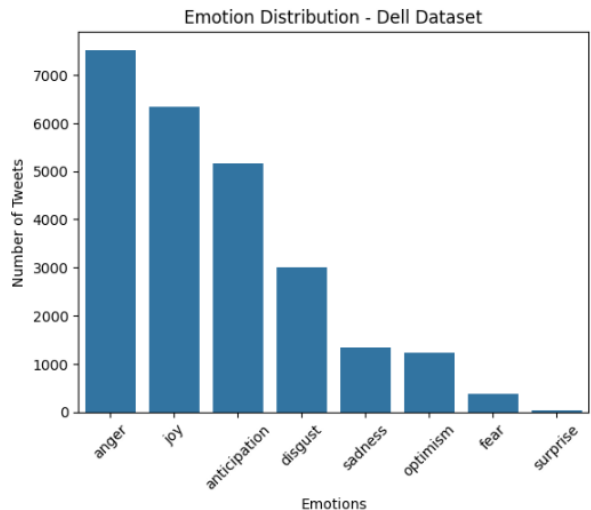


Figure 3: Emotion distribution in the Dell dataset.

The emotion distribution reflects the customer support context: *anger*, *joy* and *anticipation* dominate (Figure 3), while *optimism* remains sparse. This imbalance mirrors SemEval and reinforces the need

for label engineering.

A second key observation emerges from the temporal analysis (Figure 4): negative emotions peak during morning hours, suggesting that psychological stance varies with external factors such as work-day start or support response cycles. Acknowledging this variability, our label engineering strategy (Section 3.2) aims to extract consistent optimism-pessimism signals despite such temporal fluctuations.

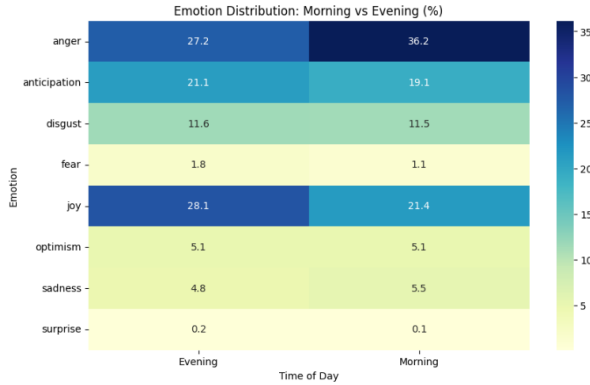


Figure 4: Temporal distribution of emotions in the Dell dataset.

3.2. Label Engineering and Class Imbalance

To bridge the gap between multi-label emotion annotations and binary optimism-pessimism classification, we propose a label engineering approach based on a weighted aggregation of emotional evidence. The core idea is to quantify how strongly each basic emotion contributes to an optimistic or pessimistic psychological stance.

Let $P(\cdot)$ denote empirical probabilities estimated from the SemEval training data. For each emotion e , we define an emotion weight as:

$$w(e) = \frac{P(\text{opt} | e)}{P(\text{opt})} - \frac{P(\text{pess} | e)}{P(\text{pess})} \quad (1)$$

where opt and pess denote the optimism and pessimism labels, respectively. This formulation captures the relative tendency of an emotion to co-occur with optimism versus pessimism, normalized by their marginal frequencies. Positive values of $w(e)$ indicate an association with optimism, while negative values indicate an association with pessimism.

For tweets without explicit optimism or pessimism annotations, we compute an aggregate optimism score:

$$S = \sum_{e \in E} w(e) \cdot L_e \quad (2)$$

where E is the set of nine basic emotions and $L_e \in \{0, 1\}$ indicates whether emotion e is present in the tweet. The resulting score S reflects the overall psychological stance expressed by the combination of emotions.

A key challenge arising from this process is **class imbalance**. To mitigate this issue and control label noise, we explore three complementary label assignment strategies:

- **A1 (Zero Threshold):** Tweets are labeled as optimistic if $S > 0$ and pessimistic otherwise. This strategy maximizes dataset size but may include ambiguous or low-confidence examples.
- **A2 (Percentile Filtering):** We introduce a neutral zone by discarding the middle 33% of the score distribution. Only high-confidence examples are retained: the top 34% are labeled as optimistic, and the bottom 33% as pessimistic. This reduces linguistic noise and improves label reliability at the cost of data volume.
- **B (Strict):** Only tweets where optimism $\neq$ pessimism are retained. This strategy yields the cleanest labels but results in a substantially smaller dataset.

To ensure fair evaluation under class imbalance, we adopt the Macro-averaged F1-score as our primary metric. Unlike accuracy, F1-Macro assigns equal importance to both classes, preventing models from achieving inflated scores by over-predicting the majority class.

3.3. Preprocessing Strategies

We compare two preprocessing regimes:

1. **Minimal Preprocessing:** We apply Unicode normalization (NFKC), URL removal, and whitespace normalization (collapsing multiple spaces and stripping leading/trailing whitespace). We intentionally preserve casing, punctuation, and

emojis, allowing Transformer models to leverage sub-word tokenization and positional embeddings to capture nuances such as sarcasm or emphasis.

2. Model-Specific Preprocessing:

- **TF-IDF Baselines:** We implement extensive pre-processing, including lowercasing, mention normalization, hashtag segmentation, and the removal of punctuation and stop-words. Finally, we apply lemmatization to reduce vocabulary dimensionality and improve the signal-to-noise ratio for linear classifiers.
- **TextCNN:** We replace usernames and URLs with special tokens (`<USER>`, `<URL>`) but preserve emojis and punctuation to maintain local syntactic and emotional context.
- **DistilBERT:** We rely on the official *distilbert-base-uncased* tokenizer which handles lowercasing and sub-word tokenization. We apply padding and truncation to a fixed length of 128 tokens.
- **BERTweet:** We rigorously follow the normalization rules from Nguyen et al. [2020], replacing mentions with `@USER` and URLs with `HTTPURL`. We preserve emojis, hashtags, and original casing to ensure full compatibility with its pre-training on 850M tweets.

3.4. Experimental Pipeline

Our approach follows three stages: computing emotion weights, creating dataset variations and evaluating models across configurations.

1. **Weight Derivation:** We compute emotion weights $w(e)$ (Table 1) from conditional probabilities of optimism and pessimism given each emotion in the SemEval training set. These weights form the mathematical foundation linking multi-label emotions to a single optimism-pessimism dimension.
2. **Dataset Transformation:** We use the derived weights to generate three versions of the SemEval dataset, reflecting different labeling strategies:
 - **Approach A1 (Full-Sample):** Utilizes continuous soft labels for all instances in the dataset.
 - **Approach A2 (Median-Thresholded):** Applies a hard threshold at the median to binarize the entire dataset.

- **Approach B (Extreme-Thresholded):** Retains only the top and bottom 30% of instances (high-confidence) to maximize the signal-to-noise ratio.

3. Model Training and Cross-Domain Transfer:

We train five model families (logistic regression, linear SVM, TextCNN, DistilBERT, BERTweet) across the three dataset variants and two preprocessing regimes, resulting in 30 experimental configurations. We then compare the performance of selected models on the Dell Tweets dataset to analyze their ability to generalize to a dataset different from the training data.

Emotion	Weight $w(e)$
anger	-0.6291
anticipation	+0.9349
disgust	-0.9371
fear	-1.1973
joy	+1.8857
love	+2.1075
sadness	-1.9827
surprise	+0.3399
trust	+2.5152

Table 1: Emotion weights derived from conditional probabilities. Positive values push towards optimism, negative towards pessimism.

3.5. Model Architectures

We implemented and tested five model families with the following configurations:

- **Linear Models (LR and SVM):** Used to establish a performance baseline. We use TF-IDF vectorization with unigrams and bi-grams (maximum 50,000 features). The regularization parameter $C \in \{0.1, 0.5, 1, 5, 10\}$ is optimized via 3-fold cross-validation to prevent overfitting on the SemEval dataset.
- **TextCNN:** An efficient neural alternative that utilizes an embedding layer (100-dimensional) followed by 1D convolutional filters of sizes 3, 4, and 5 to capture local n-grams. This architecture is particularly effective for capturing the informal and

compact syntax of tweets. Training is performed using the Adam optimizer (10^{-3}) with early stopping based on validation loss.

- **Transformers (DistilBERT and BERTweet):** Fine-tuning is performed using the HuggingFace *Trainer* API. We use a learning rate of $2 \cdot 10^{-5}$, a batch size of 16, and train for up to 5 epochs with epoch-level evaluation. We anticipate that **BERTweet** will dominate performance due to its pre-training on 850M tweets, making it highly compatible with the specific syntax and vocabulary of our Dell dataset.

4. Results

4.1. SemEval Development Performance

Figure 5 and Table 2 summarize macro-F1 on the SemEval dev split for 30 completed runs. Across all models, the percentile-based label strategy A2 is competitive and often best, suggesting that dropping the ambiguous middle-score region improves label quality. Transformers substantially outperform traditional baselines and TextCNN, with BERTweet reaching 0.9005 macro-F1 on A2.

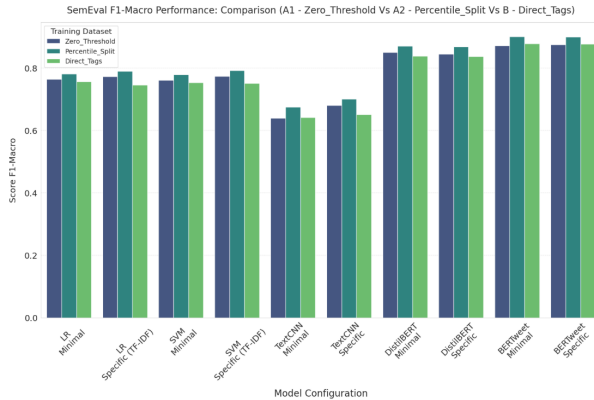


Figure 5: Macro-F1 on SemEval dev across dataset strategies (A1, A2 and B) and model configurations.

4.2. Explainability

We analyze feature importance for the TF-IDF Logistic Regression model by inspecting coefficients. Figure 6 shows that optimism correlates with positive affect words such as *love*, *happy*, *smile* and pessimism correlates with negative affect, insults

Model	Preprocessing	Train_Dataset	F1-Macro
LR	Minimal	Zero_Threshold	0.7631
LR	Minimal	Percentile_Split	0.7812
LR	Minimal	Direct_Tags	0.7558
LR	TF-IDF	Zero_Threshold	0.7724
LR	TF-IDF	Percentile_Split	0.7892
LR	TF-IDF	Direct_Tags	0.7455
SVM	Minimal	Zero_Threshold	0.7608
SVM	Minimal	Percentile_Split	0.7788
SVM	Minimal	Direct_Tags	0.7530
SVM	TF-IDF	Zero_Threshold	0.7732
SVM	TF-IDF	Percentile_Split	0.7921
SVM	TF-IDF	Direct_Tags	0.7505
TextCNN	Minimal	Zero_Threshold	0.6387
TextCNN	Minimal	Percentile_Split	0.6740
TextCNN	Minimal	Direct_Tags	0.6414
TextCNN	Specific	Zero_Threshold	0.6800
TextCNN	Specific	Percentile_Split	0.6998
TextCNN	Specific	Direct_Tags	0.6504
DistilBERT	Minimal	Zero_Threshold	0.8494
DistilBERT	Minimal	Percentile_Split	0.8696
DistilBERT	Minimal	Direct_Tags	0.8378
DistilBERT	Specific	Zero_Threshold	0.8448
DistilBERT	Specific	Percentile_Split	0.8683
DistilBERT	Specific	Direct_Tags	0.8365
BERTweet	Minimal	Zero_Threshold	0.8713
BERTweet	Minimal	Percentile_Split	0.9005
BERTweet	Minimal	Direct_Tags	0.8774
BERTweet	Specific	Zero_Threshold	0.8747
BERTweet	Specific	Percentile_Split	0.8991
BERTweet	Specific	Direct_Tags	0.8770

Table 2: Macro-F1 scores on SemEval dev set. Comparison across three labeling strategies: Zero Threshold (A1), Percentile Split (A2), and Direct Tags (B). "Specific" denotes the model-specific preprocessing.

and profanity. This matches prior observations that optimism and pessimism are related to sentiment but also include stance and intensity.

4.3. Error Analysis

For the best SemEval model (BERTweet on A2) we observe a small difference in average tweet length between correct and incorrect predictions (94.86 vs 88.63 characters). The error rate is almost identical for tweets with emoji and without emoji (9.65% vs 9.69%). Qualitative inspection suggests that many errors involve sarcasm, political context or ambiguous phrasing, for example “*why is everybody seem sp serious?*” (optimism mislabeled as pessimism) and long political tweets where stance is hard to infer without context.

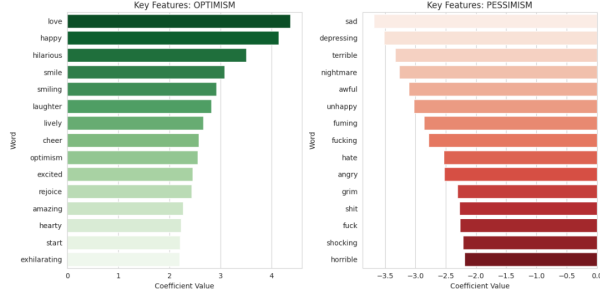


Figure 6: Top positive and negative TF-IDF Logistic Regression coefficients (A2).

5. Cross-Dataset Evaluation

To test robustness, we evaluate on the Dell tweets dataset with engineered labels. We define a score:

$$\text{score} = s \cdot \text{sentiment_score} + w(e) \cdot \text{emotion_score} \quad (3)$$

where $s \in \{-1, 0, 1\}$ maps negative, neutral and positive sentiment, e is the dataset emotion label and $w(e)$ reuses the SemEval-derived weights. Tweets with emotion *optimism* are always mapped to label 1. We create two label variants: (i) threshold 0 and (ii) percentile thresholds (33% and 66%) with middle-band drop, then we test A1 models on (i) and A2 and B models on (ii).

Figure 7 compares macro-F1 on SemEval versus the Dell dataset (Table 3). Transformers transfer strongly, reaching up to 0.9672 macro-F1 for BERTweet on A2, while linear models degrade more noticeably. TextCNN was not evaluated cross-dataset in this notebook because of its initial low performance.

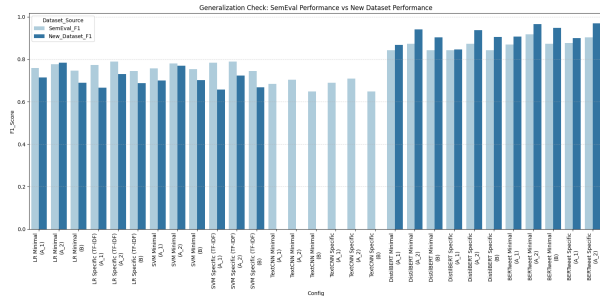


Figure 7: Generalization check: macro-F1 on SemEval dev compared with macro-F1 on the Dell dataset (engineered labels).

Model	Train_Dataset	F1 Macro
LR	Zero_Threshold	0.7041
LR	Percentile_Split	0.7718
LR	Direct_Tags	0.6968
LR	Zero_Threshold	0.6656
LR	Percentile_Split	0.7302
LR	Direct_Tags	0.6871
SVM	Zero_Threshold	0.7080
SVM	Percentile_Split	0.7718
SVM	Direct_Tags	0.7025
SVM	Zero_Threshold	0.6514
SVM	Percentile_Split	0.7253
SVM	Direct_Tags	0.6700
DistilBERT	Zero_Threshold	0.8665
DistilBERT	Percentile_Split	0.9369
DistilBERT	Direct_Tags	0.7737
DistilBERT	Zero_Threshold	0.8313
DistilBERT	Percentile_Split	0.9318
DistilBERT	Direct_Tags	0.8055
BERTweet	Zero_Threshold	0.9047
BERTweet	Percentile_Split	0.9623
BERTweet	Direct_Tags	0.9593
BERTweet	Zero_Threshold	0.9004
BERTweet	Percentile_Split	0.9672
BERTweet	Direct_Tags	0.9230

Table 3: Macro-F1 on the Dell dataset for evaluated configurations.

6. Future Work

We would like to extend the study with a true held-out test set and additional domains beyond Twitter. A natural next step is domain adaptation or knowledge distillation in the spirit of Cobeli et al. [2022]. We also plan to test sarcasm-aware features, calibration of decision thresholds and more fine-grained label definitions that separate optimism from general positivity.

7. Conclusion

We proposed label engineering strategies that transform a multi-label emotion dataset into a binary optimism versus pessimism task, then compared classic models, TextCNN and transformers under minimal and model-specific preprocessing. The percentile-based strategy A2 yielded the strongest overall results, especially for transformer models. BERTweet achieved the best SemEval dev performance (0.9005

macro-F1) and also transferred well to a second dataset with engineered labels.

Limitations

Our evaluation uses the official SemEval dev split, so reported numbers may overfit development choices. Label engineering introduces noise and may reflect biases of the underlying emotion annotations. The work focuses on English tweets, and transformer fine-tuning requires GPU resources.

Ethical Statement

Optimism and pessimism classifiers can be misused for profiling or manipulation, especially if deployed without consent. Biases can arise from demographic skews in Twitter data and from annotation subjectivity. We recommend using this work only for aggregate analyses, reporting uncertainty and auditing performance across subgroups when possible.

Acknowledgements

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